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R11T Contact Cooled Rotary Screw Air Compressor Fix Speed



RELIABILITY

- **Improved Airend** with integrated separation system
- **Improved seals** and fewer components
- **Continuous Duty** provide pulsation free air 24 hours a day, operating on a 100% duty cycle.
- **Lower RPM** -Results in lower power consumption, making it high energy efficient.
- **Hot Box Cool Box Design** – Exclusive design ensures extended life and reliability. The unique design enables air-end to achieve starting temperature as soon as possible and keep heat away from the remaining components like motor.

PRODUCTIVITY

- **Advanced Cooling** – High Efficiency, oversized combination cooler with roof mount discharge for ease of ducting.
- **Poly-V Belt Drive** – Premium drive system to minimize belt stretching and increase air output.
- **Less space** - Our space saving design gives more flexibility during installation and servicing. The small foot print, convenient location of the service panel and easy access to internal components allow the unit to be positioned in smaller areas and close to the wall.
- **Simple Diagnostics** – Visual indication of operating status, hours & fault warnings for ease of operation and reduced downtime.
- **Generous Serviceability** – Spacious design for ease of serviceability. Consumable items are extremely eased to access for quick and cost-effective preventive maintenance.
- **Ultra-Coolant** – Ingersoll Rand's Ultra-Plus provides 8,000 Hours extended coolant life, simple condensate separation and high level of biodegradability
- **Total Air System (TAS) option**– Comes with the optional tank mounted dryer model to provide the utility with the clean and dry air. Dryer is mounted on the same tank that of machine so saves a lot of space required for the installation of the separate Dryer. This option is equipped with the pre-filter to cater the clean requirement of air. Fan instead of the blower: more space to work

INGERSOLL RAND single stage contact cooled, rotary screw packaged air compressor.

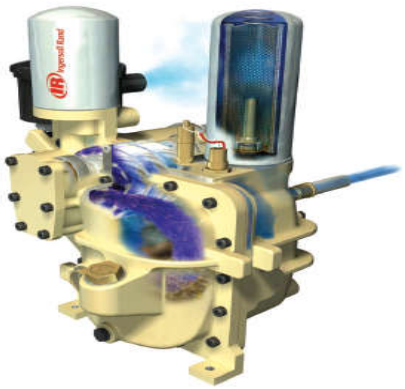
GENERAL PACKAGE DESCRIPTION:

The fully packaged contact cooled, FIX SPEED rotary screw air compressor is designed for total convenience, it is easy to install, simple to operate and will deliver rated compressed air with reliable efficiency. Operating pressure can be specified as 7.5barg, 8.5barg and 10barg as max pressure. The enclosure is designed to manage the environment of the internal components and the cooling ventilation system, at the same time as significantly reducing the sound level emitted from the machine into the installation environment. The complete package is designed to be located on an adequate flat floor without any special foundations and will operate without imparting significant vibrations to the installation.

INLET AIR FILTER:

Inlet air filtration is accomplished by a large dry type air cleaner. Minimum efficiency at 3 microns is 99.0%. This is more than suitable for the vast majority of applications:

COMPRESSION MODULE / AIR END:



Since the airend is the fundamental component in any rotary screw compressor package, reliability, performance and efficiency are determined for the most part by the design, manufacturing tolerances and assembly of the airend itself. All other elements in the compressor system are essentially support and monitoring devices included to ensure dependable service and performance.

The rotors are manufactured from AISI-1045 steel or EN 10083-2 C45+N steel. The asymmetrical helical profile is developed through a unique two-step machining process. The first step in the machining process develops the basic wrap angle profile and is a rough cut. The second and final step is a finish grinding process, which ensures a hard, true rotor surface. Rotor shafts are precision ground to tolerances within 0.5mil or 0.0005 of an inch. The rotors are paired prior to assembly into the rotor housing. The rotor housings are made of close grain high quality cast iron. After machining, the housing is dimensionally checked to ensure accuracy.

Bearing configuration used on all RT series models is the tapered rollers thrust bearing and parallel roller journal bearing. These roller bearings are able to handle all loads, radial, thrust or a combination of both. With this bearing configuration, the discharge end of the male and female rotors are each equipped with a pair of tapered roller bearings offset at opposing axis for maximum
positional stability and absorption of thrust and radial loads. The thrust bearing housing is made of a close grain cast iron.

Cylindrical roller bearings are used to carry the radial loads on the inlet end of the rotors. All bearings whether thrust or radial, are premium specification, which provide truer, harder running surfaces for both inner and outer bearing races. A double shaft seal is fitted on the main input shaft.

Coolant dams are machined at the bearing locations. This provides an area for coolant to accumulate when the compressor is shut off. Upon start-up the bearings, which are resting in coolant retained by the coolant dam is immediately lubricated, thereby assuring long life.

COOLANT RESERVOIR:

A pre-separator is fully integrated with the airend forming a single module. A pressure relief valve mounted on the housing protects the package. The coolant filler, is designed to prevent overfill the compressor, and a visual coolant level indicator is located on the side of the module. A drain point is provided at the bottom of the sump.

DRIVE MOTOR (Fixed Speed):

The main drive motor is exactly matched to the requirements of the compressor. Torque and shaft load requirements of the compressor were matched to design criteria that enable the motor to develop peak efficiency and power factor at full load of the compressor package. Standard units are supplied with IP55 motors, meeting or exceeding IE2 (IEC60034-30) efficiency rating requirements for totally enclosed fan cooled motors. Motors are having BIS approval.

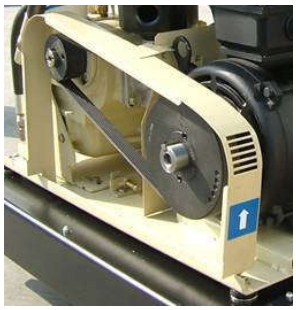
- Frame - The cast iron IEC frame motor is foot mounted. The frame design is specified to provide maximum strength and rigidity for bearing support, uniform stator/rotor gap and permanent alignment of all mating parts.

- Electrical Design - Speed, torque and operating characteristics have been designed to match the load of the compressor. Motor efficiency and power factor have been optimized. Standard motors are available in 50Hz – 400V.

- Bearings - Vacuum degassed bearings provide dependable and reliable service.

- Insulation - The Ingersoll Rand dedicated motor has a major benefit in that the maximum temperature rise the motor actually experiences is much lower than the temperature rise permitted by the design of the motor. The motor has class F insulation as standard, which means it is rated at a continuous duty for a temperature rise up to 109°C (228°F). Design limits the actual temperature rise to not exceed 89°C (192°F) in a 40°C (104°F) ambient. This is significant, since the motor life expectancy is doubled with every 10°C reduction in temperature rise. This conservative application leads to better reliability, increased life, and a much more forgiving motor under adverse conditions. All windings and leads are copper with triple coats of insulating varnish to add extra margins of protection to the motor.

BELT DRIVE ASSEMBLY:



Power transmission between the motor drive and the airend male rotor is provided by a non-stretching poly-v belt with easy to adjust belt tension control and simple access for maintenance. This setup assures performance integrity and maximum belt life. The entire drive system is shielded by protective guarding for maximum safety.

VIBRATION MOUNTS:

The airend / motor module is mounted to the sub-base on anti-vibration pads. This makes the compressor silent and smooth running.

LUBRICATION SYSTEM:

Elements of the lubrication system include:

- Coolant Filtration - The full capacity coolant filter is a 5 micron replaceable spin on element. The system contains an internal pressure relief that bypasses at 25 psid in the event that the change warning is not acknowledged.

- Coolant/Lubricant Temperature Control - A thermostatic control valve is mounted downstream of the oil cooler. The temperature sensitive element controls the flow of coolant through the oil cooler. This provides the proper injection temperature and assures fast warm-up.

- Coolant Injection - The coolant is injected through ports near the airend inlet and directed back toward the inlet cover. This ensures the best possible pre-sealing of the rotors, and an optimum mix of coolant with air. The differential pressure between the separator tank and the airend inlet induces coolant flow.

COOLANT/AIR SEPARATION:

After compression and discharge from the airend, the air heavily laden with coolant travels to the separator tank, which is part of the airend block. Entering through separator tank inlet, the air coolant mixture is directed in a circular motion around the inside of the tank. The vortex or the circular motion separates a major portion of the coolant from the air through centrifugal force.

Separator element is two-stage molded fiberglass structurally reinforced coalescing separator. There is a scavenge line which picks up coolant which has coalesced on the inside of the separator element and feeds it back to the airend inlet.

The carryover after the separator element is less than 5 ppm. Due to the conservative sizing of the separator element, there is a minimal 3 psi (0.2 bar) pressure drop. This reduces the required power to move the air through the compressor system. The separator tank is part of the airend module mounted horizontally to airend discharge. The separator vessel is protected by pressure relief safety valve mounted on the top of the tank. There is a coolant drain valve at the bottom of the tank which can be piped out to the sub-base (the pipe is delivered as compressor accessory). Sight level glass is located in front of the separator tank to allow user to check for the correct fill of the compressor with coolant. The air discharge from the separator is regulated by a minimum pressure check valve, which ensures, that when the unit is unloaded, sufficient pressure is maintained in the tank to propel the coolant through the system.

ULTRA COOLANT:

Ingersoll Rand compressors are factory filled with Ingersoll Rand Premium Coolant; a polyalkylene glycol (PAG) advanced full synthetic lubricant, providing better cooling characteristics, reduced wear through improved lubrication, and has a longer operating life than other synthetic lubricants.

COOLERS:

In standard air cooled units the discharge air is cooled to as low as within 12ºC (22ºF) of the ambient temperature, basis 38ºC (100ºF) and 40% RH. In air-cooled units cooling air which is drawn into the enclosure passes over the main motor and air-end and is then forced out through the after-cooler and coolant cooler. The coolers are of an aluminum finned tube construction and are designed to operate continuously, fully loaded, in ambient air up to 40ºC (14°F).



FAN MOTORS:

Air Mover (Blower) is mounted on non-drive end shaft of main motor. Blower is designed to offer proper airflow to cooler to take out the required heat from system.



COMPRESSOR/ CAPACITY CONTROLS (Fixed Speed):

- *The RT Series compressor includes a standard control module, which provides starting, stopping, capacity and pressure control, together with operating and safety control for the package. Operation of the compressor is very simple and user friendly.*
- *The instrument panel is mounted on the front of the compressor, directly above the starter for good visibility when either floor or receiver mounted.*
- *The control panel includes Run/Stop selector switch and reset button, "lock off" emergency stop button, pressure gauge, running hour meter. Signal lights indicate power on (green), auto restart (amber).*



As standard, this series is provided with automatic start / stop control with constant running control regulator, which allows the compressor to operate at 2 points on the capacity curve. The first is 100% full-flow. The second is no-flow.

The online / offline auxiliary control is a constant running mode of operation and should be selected when load conditions require.

STARTER:

11KW has start delta starter and 7KW has DOL starter. The standard compressor has an integrally mounted starter box with an openable door panel. It contains all required electrical components. The star-delta starter is used to reduce the current inrush on starting (only for 11kW case). The control relays operate at 110V AC and the control circuit is protected by miniature circuit breakers or fuses. All electrical equipment is designed to confirm to the applicable local electrical codes.

ENCLOSURE:

The package enclosure is carefully designed to provide effective sound emission control and suppression, while retaining easy access for maintenance and eventual refurbishment. The front door lifts off if required to provide easy access to all routine maintenance points. This door provides easy access to carry out the following maintenance procedures.

- Check and top up coolant
- Check intake filter condition
- Change intake filter
- Change coolant filter
- Change separator cartridge
- Service Intake valve
- Check or adjust constant running valve
- Check shuttle valve
- Drain & refill coolant
- Adjust belt tension
- Set and adjust load and unload operating pressures





Receiver Tank (Optional):

- 272L capacity receiver tank can be supplied with unit as optional.
- Design of receiver is as per local standard.
- Construction of material is as per IS2062.



Dryer/TAS (Optional):



A TAS option is available consisting of a refrigerated dryer with a High Efficiency (Grade H) filter mounted in-line before the dryer. The dryer has its own power supply and runs independently to the compressor operation. Delivered air quality of the package with the TAS module installed is to meet ISO 8573.1 class 1-5-1 (Dirt – 1 micron, Dew Point 7 degree C, Oil 0.01 mg/m3) air quality requirements at 25°C (77 °F), 60% RH. The dryer is designed such that it will operate and provide dew point suppression across the normal operating conditions of the compressor (2°C-40°C/37°F -104°F ambient). The system consists of a refrigerant compressor, condenser, condenser fan, capillary, refrigerant filter, evaporator, moisture drain, and hot gas bypass valve. The hot gas bypass is used to assist with Variable Speed operation across the flow range. R134A refrigerant is used to ensure optimum performance.

Factory tested

Every compressor undergoes rigorous computerised factory validation tests according to ISO1217:1996, Annex C, to ensure that it delivers its promised performance under a wide range of conditions.

Technical Data:

Technical data at reference conditions-

Model	R7T-10-272
Operating pressure	10 barg
CFM	33CFM
Electric Motor	7kW/10HP
Low sound enclosure:	69 dB(A)
Dimension:	826x759x782
Weight:	261 kgs.
Air Receiver:	272Lts

Description:

- High efficiency electric motors IE2 IP54, 400V/3/50Hz
- After cooler with water separator
- Star/delta start up
- Oil filled in factory

SWASTIK MACHINE TOOLS
SINCE 1982

Commercial Offer

Sr.	Description	Price INR.	Qty.	Disc.%	Discounted Amt.
1	Air Cooled, Single Stage, Lubricated Screw Compressor Fix speed, Tank Mounted Model- R7T-10-272	4,55,000	1 Nos.	39%	2,77,550

Terms & Conditions:.

Prices: - Ex-Works Chakan,

Payment: - 40% Advance & balance Against Performa Invoice Prior to Dispatch.

Delivery:- Within 3 weeks from the receipt of confirmed order.

Tax: - 18% GST will be applicable.

Other Charges: - Local Freight, Insurance will be charged extra at your account.

Installation & Commissioning: - Will be free of cost within local boundaries.

Piping & Fitting: - Will be at your scope.

Warranty: - Twelve Months from the date of commissioning or eighteen Months from the date of dispatch, whichever occurs earlier. However the warranty terminates if the unit/system is exported from India.

VALIDITY:- 30Days.

For SWASTIK MACHINE TOOLS,

Ref.by Mr.Dilip Vasmate